Software Requirement Specifications

SkillBridge-AI Powered Career Guidance System

Submitted by

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Meeting Details

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Summary

SkillBridge is an AI-powered career guidance platform designed to help students identify suitable career paths based on their skills, interests, academic strengths, and personality traits. This SRS document defines the complete functional and non-functional requirements of the system, its architecture, intended users, and system behavior. The document outlines how SkillBridge uses intelligent algorithms to analyze user input and generate personalized career recommendations. It also discusses the proposed technologies, use cases, and constraints associated with the system.

Table of Contents

1. Introduction

1.1. SkillBridge is an AI-powered career guidance platform designed to help students identify suitable career paths based on their skills, interests, academic strengths, and personality traits. This SRS document defines the complete functional and non-functional requirements of the system, its architecture, intended users, and system behavior. The document outlines how SkillBridge uses intelligent algorithms to analyze user input and generate personalized career recommendations. It also discusses the proposed technologies, use cases, and constraints associated with the system.

2. Purpose

The main purpose of SkillBridge is to:

- Provide personalized career recommendations based on student data.
- Help students identify strengths, academic capabilities, and interest areas.
- Act as a digital counselor that bridges the gap between student potential and real-world career opportunities.
- Support decision-making by offering detailed insights into career fields

2.1. Scope

The system includes:

✓ Student registration & login
✓ Career assessment questionnaire
✓ AI-based analysis of user input
✓ Career recommendation engine
✓ Display of detailed career information
✓ Admin panel to manage data

The system does **not** include:

✗ Real-time chat with human counselors
✗ Job placement or internship matching
✗ External admission applications

2.2. Product Perspective

SkillBridge is a standalone web-based system. The recommendation module is central to the platform and interacts with:

- Database for student profiles
- ML model for career prediction
- Admin backend for content management

2.3. User Characteristics

Student:

- Basic computer and internet knowledge
- Able to answer assessment questions
- Age 15–25

Admin:

- Responsible for managing career database
- Knowledge of system operations

2.4. Similar apps and systems/Literature Review

Existing career guidance apps include:

CareerGuide, **MyNextMove**, **Shine Learning**
These platforms offer career suggestions but often lack:

- Deep personalization
- AI-driven predictive models
- User-friendly structured assessments
- SkillBridge improves these shortcomings by integrating data-driven algorithms, psychological assessment components, and interactive UI.

2.5. Proposed Technologies

- **Frontend:** React / HTML / CSS / JavaScript
- **Backend:** Django / Node.js
- **Database:** MySQL / Firebase
- **AI/ML:** Python (Decision Trees, Recommendation Models)
- **Hosting:** Cloud-based (AWS / Firebase)

2. Requirements

2.1.

SkillBridge provides students with a structured career assessment process. Students enter personal data, interests, academic performance, and personality traits. The system analyzes this data and generates personalized career paths. Admin manages database, verifies system functionality, and updates career fields.

2.2. Function Requirements

There are following function requirements :

2.2.1. Sign Up

FR001 – Sign Up

Name: FR001
Purpose: To register new users and allow them to access the system.
User(s): Student
Input:

- Full Name
- Email
- Password (min 8 characters, must include numbers)
- Phone Number
- Academic Level

Output:

- Successful creation of a new student account

FR002 – Login

Purpose: Allow registered users to access the system.
Input: Email, Password
Output: Redirect to dashboard

FR003 – Career Assessment Questionnaire

Purpose: Collect student's interests, skills, academic scores, and personality traits.
Input: Multiple-choice and rating-scale answers
Output: Completed dataset for analysis

FR004 – AI-Based Career Recommendation

Purpose: Generate personalized career suggestions.
Input: Data from questionnaire + academic inputs
Output: Top recommended career paths (e.g., Engineering, Medicine, IT, Design, Management)

FR005 – Career Details Display

Purpose: Provide detailed information for each recommended career.
Output Includes:

- Job roles
- Required qualifications
- Skills to develop
- Market trends

FR006 – Admin Career Management

Purpose: Admin updates career list and system database.
User: Admin

Input: Career information fields
Output: Updated career data visible to students

2.3. Non-Functional Requirements

NFR001 – Performance:

- The system must generate recommendations within 3–5 seconds.

NFR002 – Availability:

- The system should be available 24/7 with 99% uptime.

NFR003 – Usability:

- Interface must be easy for students aged 15–25.

NFR004 – Security:

- Passwords must be encrypted.
- User data must be protected by standard security protocols.

NFR005 – Scalability:

- System must handle up to 10,000 concurrent users.

NFR006 – Reliability:

- Recommendations must be consistently accurate and derived from validated models.

3. Use Cases and Flow of Processes

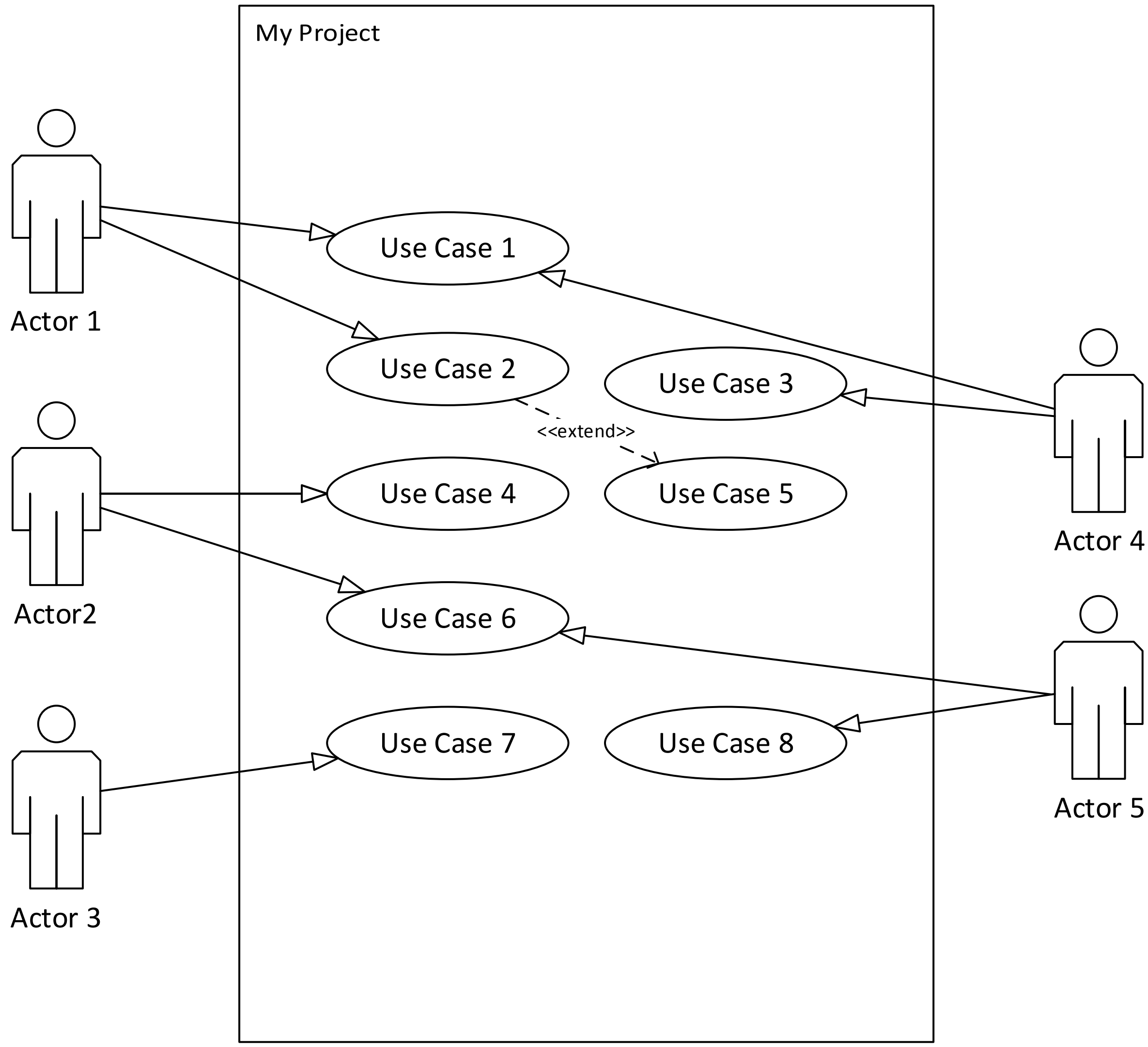

Figure 1: System Level Use Case Diagram

3.1. Use Case 1

ID	UC001
Name	Sign Up

Description	User registers to use the SkillBridge platform.
Requirement(s)	FR001
Actor(s)	Admin, student
Precondition	User is not already registered.
Postcondition	User account created and stored in the system.

Basic Flow	***Basic Flow***
	• User opens registration page.
	• User enters required information.
	• System validates inputs:
	  • Valid email format
	  • Password meets security criteria
	  • Phone number is 11 digits
	• On success, system creates a new user profile.
	• System redirects to login page.
	Alternative Flow
	• If email already exists → system shows error message.
	• If input invalid → system prompts corrections.
	Exceptions
	• Server offline
	• Database error
	• Network connection failure

4. References

- Decision Tree Algorithms for Career Prediction – IEEE Access
- User Modeling in AI Recommendation Systems – ACM
- Career Assessment Models – Psychological Review